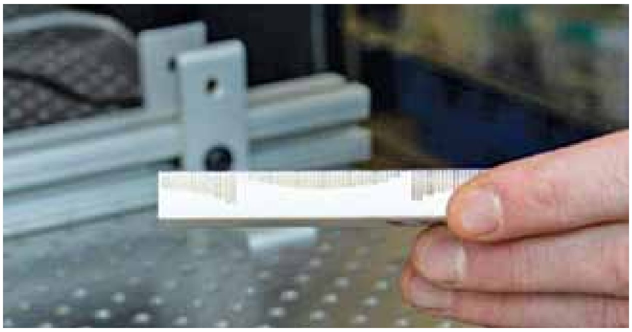
that occur within the batteries using the SM beamline at the Canadian Light Source (CLS)—located at the University of Saskatchewan (UofS). The team is interested in identifying in real time the flaws that occur, which would allow them to measure the complex relationship between the battery's materials, its shape, and the chemical reactions that occur within. Reaching this goal would allow researchers to proactively address design failures. More efficient batteries can help reduce waste and help us transition to a greener grid.

Objects steered with the help of ultrasonic waves

Researchers at University of Minnesota have developed a way to freely manipulate objects with the help of ultrasound waves. Their method can move larger objects using the principles of metamaterial physics. Metamaterials are



By placing a metamaterial pattern on the surface of an object, the University of Minnesota researchers were able to use sound to steer it in a certain direction without physically touching it (Credit: Olivia Hultgren)

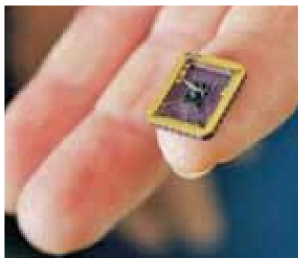
artificially designed materials that can interact with waves like light and sound. By placing a metamaterial pattern on the surface of an object, the researchers were able to use sound to steer it in a certain direction without physically touching it. When these tiny patterns are placed on the surface of the objects, one can basically reflect the sound in any direction. And in doing that, the user can control the acoustic force that is exerted on an object. Using this technique, the researchers can not only move an object forward but also pull it toward a source—not too dissimilar from the "force" described in Star Wars.

A new system devised to teach deep learning more efficiently

Researchers at Texas A&M University, Rain Neuromorphics, and Sandia National Laboratories have devised a new system for training deep learning models more efficiently and on a larger scale. This approach could reduce the carbon footprint and financial costs associated with the training of AI models, thus making their large-scale implementation easier and more sustainable. To do this, they had to overcome two key limitations of current AI training practices. The first challenge was the use of inefficient hardware systems based on GPUs, whereas the second entails the use of ineffective and math-heavy software tools, specifically utilising the so-called backpropagation algorithm. Researchers developed a new co-optimised learning algorithm that exploits the hardware parallelism of memristor crossbars. This algorithm, inspired by the differences in neuronal activity observed in neuroscience studies, is tolerant to errors and replicates the brain's ability to learn even from sparse, poorly defined, and "noisy" information.

Unique nano chips formed using silicon carbide chips

Researchers at the Georgia Institute of Technology have created a modified form of epigraphene on a silicon carbide crystal substrate to create a new nanoelectronics platform. They produced unique silicon carbide chips from electronics-grade silicon carbide crystals. They used electron beam lithography, a method commonly used in microelectronics, to carve the graphene nanostructures and weld their edges to the silicon carbide chips. This process mechanically stabilises and seals the graphene's edges, which would otherwise react with oxygen and other gases that might interfere with the motion of the charges along the edge. Results showed that the graphene edge states were similar to photons in an optical fibre that can travel over large distances without scattering. They found that the charges traveled for tens of thousands of nanometres along the edge before scattering.



Claire Berger, physics professor at Georgia Tech, holds the team's graphene device grown on a silicon carbide substrate chip (Credit: Jess Hunt-Ralston, Georgia Tech)

Eco-friendly wirelessly powered IoT sensors on the way

KAUST alumni Kalaivanan Loganathan, with Thomas Anthopoulos and coworkers, assessed the viability of various large-area electronic technologies and their potential to deliver eco-friendly, wirelessly powered IoT sensors. He has developed a range of RF electronic components, including metal-oxide and organic polymer based semiconductor devices known as Schottky diodes. "These devices are crucial components in wireless energy harvesters and ultimately dictate the performance and cost of the sensor nodes," Loganathan says. Key contributions from the KAUST team include scalable methods for manufacturing RF diodes to harvest energy reaching the 5G/6G frequency range. Such